Final Workflow: APAAR-Based Student Data Management & Verification Platform

Smart India Hackathon 2026 | Deadline: 28 October, 11:59 PM | Build time: 2 days

How to use this document: every section first gives the plan in short form, then "In simple words" explains it. Follow the plan in order. The Compliance Manager checks every step against the Problem Statement (PS).

---

1. What We Are Building

A central platform where AICTE-approved colleges submit student data. The system checks the data automatically, tracks every problem until it is fixed, shows live dashboards, and lets authorized people verify a student record. The APAAR ID is the common identifier for each student.

We ARE building	We are NOT building
A validation, monitoring and verification layer for AICTE	A replacement for college ERPs
An extension module that plugs into AICTE and institutional systems	A new APAAR ID system (APAAR IDs are issued by the government)
A tool that finds problems and tracks their fixing	A new AICTE approval portal

In simple words: Today AICTE receives student data in many different ways, so the data is hard to trust. We create one common place for it. Colleges keep their own software. We only standardize the data when it enters our system.

---

2. Phase 0: Scope Lock (Before Any Code)

Goal: all six people agree on what we are building. Otherwise their work will not connect later.

2.1 Freeze the two datasets

Dataset	What it contains	Who controls it
Approval Master	Institution, campus, programme, academic year, approved intake, supernumerary / NRI / OCI / FN seat limits	AICTE
Student Enrolment Repository	APAAR ID, PRN / local student ID, institution, programme, year, category, NRI / OCI / FN flags, admission details, status, source	Institutions submit; platform stores accepted records

In simple words: The Approval Master is the rulebook (how many students each college may admit). The Repository is the actual data (the students colleges send). Our system compares the second against the first to catch problems like "more students than approved seats". Colleges cannot change the rulebook.

2.2 Freeze the status flow

`UPLOADED → VALIDATING → ACTION_REQUIRED or PENDING_REVIEW → READY_TO_ACCEPT → ACCEPTED → VERIFIED` (and `REJECTED`)

Status	Meaning
UPLOADED	We received the record
VALIDATING	Checks are running
ACTION_REQUIRED	The college must fix something
PENDING_REVIEW	An AICTE reviewer must look at it
READY_TO_ACCEPT	All checks passed
ACCEPTED	Stored in the central repository
VERIFIED	Confirmed by an authorized source
REJECTED	Not accepted, with the reason recorded

In simple words: Think of tracking a parcel. Two rules to remember:

- ACCEPTED is not the same as VERIFIED. Accepted means "our checks passed and it is stored". Verified means "an official source confirmed it".
- A 12-digit APAAR ID that looks correct is not proof that the ID is real.

2.3 Freeze the three roles

Role	What they can do
Institution	See and edit only their own college's data
AICTE	See all data for monitoring
External Verifier (employer, regulator)	Check one record only, with consent. Cannot browse the database

2.4 Prepare synthetic data (50 to 100 records)

Use made-up students only, never real ones. Include deliberate faults to prove our checks work:

Planted fault	Check it tests
Missing APAAR ID	Missing-ID check
Wrong format	12-digit format check
Duplicate inside the same batch	Duplicate detection
Same student in two institutions	Cross-institution conflict
125 enrolled vs 120 approved intake	Excess-enrolment check
NRI / FN seats over the limit	Seat-category limits
Wrong programme mapping	Course-to-college matching

2.5 Set up the team

- One GitHub repository for all code.
- A decision log: a written list of what we decided and why, so we do not repeat arguments.
- One accountable owner per module, so two people do not change the same part.
- A post-MVP list: a parking lot for good ideas we will not build now.

2.6 The Compliance Gate

Ask these three questions for every new suggestion:

1. Which PS requirement does it fulfil?
2. Is it required or optional?

3. Does it risk the end-to-end flow?

If there is no clear PS requirement, it goes to the post-MVP list. No debate.

In simple words: This is the Compliance Manager's main job. It stops the team from drifting away from the PS, which our senior warned about.

---

3. Day 1: Core Flow (Upload, Validate, Show)

Goal: a college uploads a file, the system checks it, and shows exactly what is wrong.

Step 1: Data model and API contract

Plan: Design the database tables, the request/response formats and the screen flow. Then freeze the contract.

In simple words: The data model is the design of our tables. The API contract is the agreement on what goes into each API and what comes back. If it keeps changing, backend and frontend will never connect.

Step 2: Ingestion

Plan: CSV/Excel upload, field mapping to a standard format, batch ID, visible batch status, Approval Master pre-loaded.

In simple words: Colleges name columns differently ("Roll No", "PRN", "Student ID"). We convert all of them into one standard format. Every upload gets a batch number so we can track it, and the user can see its progress.

Step 3: Validation engine

Plan: Rule-based, explainable checks:

Check	What it asks
Required fields	Is anything important missing?
APAAR format	Is it exactly 12 digits?
Duplicate in batch	Is the same APAAR ID repeated in the same file?
Duplicate in repository	Does this ID already exist in a conflicting record?
Cross-institution conflict	Is the same student actively enrolled in two colleges?

Check	What it asks
Course / institution mapping	Does this course exist at this college in the Approval Master?
Intake check	Are there more students than approved seats? Compare per institution, programme, year and seat category
Data consistency	Are dates valid and fields consistent with each other?

In simple words: These are plain if/else rules, not AI, so government officers can easily understand them. For the intake check, we never compare a whole college's student count against one course's seats. A duplicate across colleges is flagged for review, not called fraud, because transfers and withdrawals are legitimate.

Step 4: Issue workbench UI

Plan: A table of every problem found. Filters: institution, course, year, issue type, severity, status. Each row shows the reason and evidence.

In simple words: This is more useful than a plain spreadsheet because users can act on each problem.

Step 5: Integration test

Plan: Run one clean batch and one faulty batch. Every planted fault must show up as flagged.

In simple words: This proves the whole path (upload, check, display) works.

> **DAY 1 EXIT RULE:** If a faulty batch does not produce correct flagged rows with reasons, nothing else starts (no dashboards, API or verification). Everything else depends on this.

---

4. Day 2: Close Every Loop

Goal: finish the remaining requirements and complete the correction loop.

Step 6: Correction loop (highest priority)

Plan: Every issue has an owner, a required action, a due date (configurable, default 7 days, labelled as a prototype setting) and an escalation path. The institution fixes the row, resubmits, and the system revalidates automatically. Unresolved or overdue issues escalate to the AICTE reviewer.

In simple words: In SIH Round 1, marks were cut because problems were flagged but nothing happened after that. This step fixes exactly that. The cycle is: flag, assign, fix, resubmit, recheck, escalate if still unresolved. Nothing is silently dropped or marked verified.

Step 7: Dashboards

Plan: Two views. Institution view shows only its own data. AICTE view shows intake vs enrolment, utilisation, seat-category counts, pending and overdue anomalies. Numbers update when records are accepted.

In simple words: When a record is corrected and accepted, the numbers change. This is how we show "real-time".

Step 8: API and security

Plan: Bulk upload and update endpoints with a sample request. Authentication, role-based access, audit log (who, what, when, change), consent record.

In simple words:

- The API lets a college's software send data directly instead of uploading a file.
- Authentication means users must log in.
- Role-based access means each role sees only what it is allowed to see.
- The audit log records who did what and when.
- The consent record stores that a student allowed their record to be checked.

Step 9: Verification and certificate

Plan: QR code / unique link leading to a verify page that shows only permitted fields and handles invalid or revoked records. A printable certificate preview showing both APAAR ID and institutional student ID. Demo mode and production-adapter mode clearly labelled.

In simple words: Scanning the QR opens a page with limited information. A fake or revoked record shows a clear failure message. The certificate prints the APAAR ID next to the student ID, which is a PS requirement.

Step 10: Reports

Plan: Downloadable intake and anomaly reports (CSV, and PDF if time allows).

In simple words: These serve policy-making, accreditation and monitoring.

Step 11: Full test and feature freeze

Plan: Run all acceptance tests, document endpoints and limitations, capture screenshots. After this, no new features.

---

5. Resolution Map: Who Fixes What

This answers the Round 1 question: "after a flag is raised, who fixes it?"

Issue	Responsible	Resolution path
Missing / invalid APAAR ID	Institution data officer	Correct and resubmit, or keep the record pending
Wrong course mapping	Data officer + master-data admin	Check against the Approval Master, then fix the source
Intake exceeded	Admissions officer, then AICTE reviewer	Reconcile, check supernumerary rules, escalate if unresolved
Cross-institution duplicate	Both institutions + reviewer	Confirm transfer/withdrawal evidence (flag for review, not fraud)
Inconsistent data	Data officer	Fix fields and revalidate
Verification service down	Integration admin	Retry or manual check; record stays pending

Important limit: The platform finds problems and tracks their fixing. It never edits approved intake, never generates APAAR IDs, and never grants AICTE exceptions itself.

---

6. Acceptance Tests (All on Synthetic Data)

A feature counts as done only when its test passes, not when the code is written.

Test	Expected result
Missing APAAR ID	Issue raised

Test	Expected result
Bad format	Issue raised
Duplicate in batch	Rows flagged
Cross-institution conflict	Review case created
125 vs 120 intake	Potential excess flagged
Correction submitted	Revalidates, status updates, dashboard refreshes
Institution A opens institution B's data	Denied and logged
Valid, invalid and revoked QR	Correct result for each
Mixed valid/invalid batch	Invalid rows are not counted as accepted
Certificate generated	Shows both APAAR ID and student ID
Fresh clone, following only the README	Runs with no help

---

7. The Demo Story

Show one continuous story instead of many separate screens:

1. Upload a faulty batch
2. Anomaly report appears
3. Institution corrects the problems
4. System revalidates
5. Records are accepted into the central repository
6. AICTE dashboard updates
7. An authorized verifier checks one record
8. Certificate shows both IDs

In simple words: One complete story is more convincing than ten disconnected screens. Practise it several times.

---

8. Requirement Checklist (PS Requirement to Demo Evidence)

#	PS requirement	Evidence in demo
1	Secure upload / update / API	CSV upload + working API call
2	Required data fields	Field-level view + Approval Master
3	Automated validation	All checks fire on the faulty batch
4	Real-time dashboards	Correct a record, metrics refresh
5	Authorized verification	Restricted verification request
6	Verifiable student records	QR / unique link verify page
7	Integration capability	Documented REST API + CSV fallback
8	Security and privacy	Allowed vs denied access, audit log, consent
9	Reports	Downloadable intake and anomaly report
10	APAAR ID on certificates	Certificate preview with both IDs

---

9. Do Not Build

Item	Why not
Blockchain	Too complex; government staff are not cyber experts
ML anomaly scoring	Rule-based checks meet the PS and are easier to explain
Mobile app	Not asked for in the PS
Multi-ERP connectors	We offer an API and CSV fallback instead
Vendor marketplace	The college, our system and the AICTE reviewer already cover all roles
National-scale claims	We only claim what we can show with sample data

---

10. USPs (Only After All Tests Pass)

USPs must make existing requirements work better, not add unrelated features.

1. **Auto-escalation:** an issue unresolved past its due date automatically goes to the AICTE reviewer.
2. **Issue prioritisation:** the most serious, highest-impact issues are shown first.
3. **Pre-validation check:** a college can check its file for errors before submitting.

---

11. Honesty Rules

Judges respect honesty more than exaggerated claims.

- We do not have real DigiLocker or APAAR access, so we label it simulated / demo mode and show how the real connection would plug in.
- State clearly: a correct-looking 12-digit ID is format-checked, not identity-verified.
- State that the reference data is a sample and the student records are synthetic.
- Claim a feature as done only if it has been tested.

---

12. Compliance Manager Routine

When	What to do
Every 3 to 4 hours	Go through the 10 PS requirements and mark each: done, partly done, or not started
Every blocker	Record it, give it one named owner, decide how it will be solved. Never leave a problem without an owner (the Round 1 lesson)
Every new suggestion	Ask the three gate questions, then build it or log it as post-MVP
Final gate	No feature is called complete unless tested. No integration is called real unless it is real